PROJECT REPORT

Student Course Management System

JAVA CONSOLE APPLICATION | SEMESTER COURSE TRACKER

What the system does

This Java application helps a student keep an organized list of semester courses. Each record contains a course code, title, and positive credit-unit value. A numbered console menu provides the main course-management actions.

How the main features work

- **Add a course**
  Enter a code, title, and whole-number units. The program trims text, normalizes codes to uppercase, and rejects blank fields, invalid units, or duplicate codes.
- **View and search**
  View prints the saved-in-memory courses in a readable table. Search matches by course code without depending on letter casing; its recursive method checks the list one course at a time.
- **Calculate units**
  The total-units option iterates over all recorded courses and adds their credit units.
- **Save and load**
  Save writes one course per line to courses.txt. Load validates all rows first, then replaces the current list only when the file is valid.
- **Menu and errors**
  A loop keeps the menu available until Exit is chosen. Invalid selections, non-numeric units, bad file data, and file-access problems are reported to the user.

Program in action

Sample session: adding two courses, viewing the list, and totaling units.

Student Course Management System | Java

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COS 201         Programming I                   3
MTH 101         Mathematics                     2

=== Student Course Management System ===
1. Add Course
2. View All Courses
3. Search Course by Code
4. Compute Total Units
5. Save to File
6. Load from File
7. Exit Program
Choose an option (1-7): 4
Total units: 5

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Program design and validation

Course, CourseManager, and CourseManagementApp separate course data, course operations, and the menu interface. The recursive search stops when it finds a matching code or reaches the end of the list. File loading builds and validates a temporary list before changing existing records, so a malformed file cannot leave a partial update.

Testing

The standalone Java test program checks adding courses, normalized search, unit totals, saving and loading, duplicate-code rejection, and preservation of existing data after a failed load. The application and tests compile without external libraries.